

LAWRENCEVILLE-VINCENNES, IL, USA

WMO: 725342

Lat: 38.764N

Lon: 87.606W

Elev: 131

StdP: 99.76

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 13809

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-11.5	-19.6	0.7	-13.1	-17.1	0.8	-11.0	12.6	1.8	11.6	4.8	3.2	320	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.3	24.2	33.0	24.0	32.1	23.6	26.4	32.0	25.6	30.9	24.9	29.8	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.2	29.5	24.0	19.2	28.6	23.2	18.3	27.5	82.6	31.9	79.2	30.9	76.0	29.7	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.4	8.4	-18.9	36.3	2.8	2.0	-21.0	37.7	-22.7	38.8	-24.3	39.9	-26.3	41.3
			-19.3	27.8	2.7	1.0	-21.3	28.5	-22.9	29.1	-24.5	29.7	-26.5	30.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-0.4	1.8	7.2	13.7	19.3	23.8	25.2	24.5	21.1	14.3	7.5	1.8	(6)
(7)		DBStd	10.33	6.36	6.05	5.91	5.00	4.51	3.25	2.82	2.73	3.84	5.06	5.20	5.91	(7)
(8)		HDD10.0	1094	327	236	126	21	1	0	0	0	15	108	261		(8)
(9)		HDD18.3	2595	582	463	346	153	46	3	0	1	146	326	513		(9)
(10)		CDD10.0	2323	4	7	39	131	288	413	470	449	333	149	33	7	(10)
(11)		CDD18.3	782	0	0	2	13	74	166	212	191	101	22	1	0	(11)
(12)		CDH23.3	7448	0	1	17	129	652	1576	2111	1773	966	218	6	0	(12)
(13)		CDH26.7	2965	0	0	2	21	216	646	907	723	386	64	1	0	(13)
(14)	Wind	WSAvg	3.8	4.6	4.5	4.6	4.6	3.9	3.2	2.7	2.5	2.8	3.6	4.0	4.1	(14)
(15)	Precipitation	PrecAvg	1106	68	62	100	104	133	102	110	96	77	79	101	83	(15)
(16)		PrecMax	1526	206	137	272	246	315	222	278	300	215	247	284	206	(16)
(17)		PrecMin	806	12	12	18	26	34	11	6	16	0	5	3	14	(17)
(18)		PrecStd	168	45	32	57	53	74	52	63	57	49	56	54	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	20.4	25.9	29.0	32.7	35.1	36.7	36.1	34.9	31.3	23.8	18.8	(19)
(20)			MCWB	15.4	15.6	17.8	17.9	21.9	23.6	26.0	25.2	22.6	21.5	16.2	15.9	(20)
(21)		2%	DB	15.0	16.9	22.5	26.9	31.1	33.4	34.1	33.8	32.8	28.2	21.1	15.8	(21)
(22)			MCWB	12.5	13.0	15.8	18.4	21.5	23.7	25.2	24.3	22.6	19.5	15.6	13.8	(22)
(23)		5%	DB	12.2	13.9	19.6	24.6	29.1	32.3	32.7	32.3	31.1	25.9	18.2	12.8	(23)
(24)			MCWB	10.5	10.8	14.2	16.9	20.6	23.3	24.9	24.1	22.1	18.2	14.2	10.6	(24)
(25)		10%	DB	8.9	11.2	17.1	22.4	27.3	30.7	31.3	30.8	28.8	22.9	16.2	10.4	(25)
(26)			MCWB	6.7	8.3	12.7	15.9	20.1	22.6	24.1	23.4	20.7	17.0	12.2	8.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	16.6	18.6	21.3	24.5	26.3	27.6	27.2	25.1	22.9	19.0	16.7	(27)
(28)			MCDWB	17.4	19.7	24.6	26.3	29.5	32.2	33.0	32.4	31.4	29.0	22.4	18.4	(28)
(29)		2%	WB	13.1	14.0	16.9	19.6	23.1	25.3	26.7	26.1	24.0	21.3	16.7	13.9	(29)
(30)			MCDWB	14.6	16.0	21.3	24.4	28.1	30.9	32.3	31.5	29.8	26.1	19.3	15.3	(30)
(31)		5%	WB	10.6	11.3	15.2	18.3	22.1	24.5	25.9	25.2	23.1	19.7	14.9	11.0	(31)
(32)			MCDWB	12.0	13.7	18.8	22.3	27.0	29.8	31.2	30.4	28.7	23.8	18.1	12.5	(32)
(33)		10%	WB	6.7	8.0	13.2	17.0	21.1	23.7	25.2	24.4	22.2	18.1	12.6	8.3	(33)
(34)			MCDWB	8.4	10.4	16.5	21.0	25.4	28.7	30.1	28.9	26.7	21.9	15.7	9.8	(34)
(35)	Mean Daily Temperature Range	MDBR		8.7	9.3	10.6	11.5	10.8	10.9	10.6	11.2	12.9	12.2	10.4	8.2	(35)
(36)		5% DB	MCDBR	11.5	13.4	14.0	14.2	13.0	12.7	11.9	12.8	14.7	14.9	13.6	10.9	(36)
(37)			MCWBR	10.0	10.4	8.7	7.5	5.5	4.9	4.8	5.4	5.7	7.2	9.1	9.1	(37)
(38)		5% WB	MCDWB	11.0	12.1	12.4	11.4	10.8	10.7	10.6	10.8	12.2	12.6	11.8	9.7	(38)
(39)			MCWBR	10.3	10.4	8.4	7.5	5.4	4.8	5.1	5.4	5.5	6.7	9.2	9.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.312	0.324	0.360	0.409	0.445	0.449	0.468	0.475	0.423	0.362	0.323	0.308	(40)
(41)		taud		2.448	2.394	2.365	2.229	2.179	2.199	2.149	2.116	2.251	2.422	2.476	2.466	(41)
(42)		Ebn at Noon		864	900	898	868	840	834	814	798	820	845	843	841	(42)
(43)		Edh at Noon		82	98	112	137	147	144	150	152	125	95	79	75	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.89	2.67	3.71	4.88	5.49	6.33	6.12	5.60	4.65	3.35	2.26	1.62	(44)
(45)		RadStd		0.21	0.29	0.34	0.29	0.48	0.40	0.39	0.32	0.33	0.35	0.24	0.16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page